

CALORIMETRY USING A BOMB CALORIMETER

1. Goal

In this experiment, the enthalpies of combustion of maleic and fumaric acid are to be determined. Based on the results, it is possible to draw conclusions about their enthalpy of conversion. To achieve this, however, the heat capacity of a bomb calorimeter must be determined first.

2. Theoretical background

2.1 Heat transactions

The following reaction is investigated:



It takes place within the calorimeter. Reaction (I) is associated with the absorption or release of a quantity of heat ΔQ per formula turnover λ . For constant temperature & pressure *or* constant temperature & volume, eq. (1) *or* eq. (2) apply:

$$\Delta Q = \lambda \sum v_i H_i \quad p, T = \text{const.} \quad (1)$$

OR

$$\Delta Q = \lambda \sum v_i U_i \quad V, T = \text{const.} \quad (2)$$

with: v_i = stoichiometric coefficients
 H_i *or* U_i = partial molar enthalpy *or* energy of the reactants

The following reaction types can be distinguished:

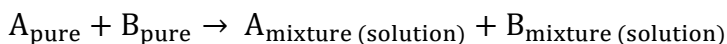
- a. Chemical reactions in a narrow sense:

$$\Delta Q = \text{heat of reaction } \Delta_r Q$$

- b. Phase transitions:

$$\Delta Q = \text{heats of fusion, condensation, sublimation, transformation}$$

- c. Mixing processes (dissolution processes):



$$\Delta Q = \text{Heat of mixing (heat of dissolution) } \Delta_m Q$$

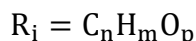
Refer to the referenced literature [1] for information on the terminology used in connection with heat transactions and energy conversions.

What does Hess's law state (see [1], chapter 2C.1)? What does the "enthalpy of formation" of a compound mean (see [1], chapter 2C.2)? What does Kirchhoff's law state (see [1], chapter 2C.3)?

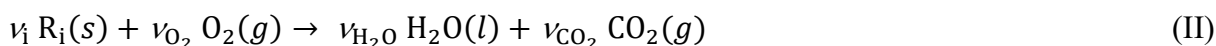
These questions are to be answered along with the exercises (see chapter 6)!

2.2 Enthalpy of combustion

In the experiment at hand, the enthalpy of *reaction* of a chemical reaction is to be determined indirectly from the enthalpy of *combustion* of the reactants according to Hess' law. The reactants R_i are solid, pure organic substances of the following composition:



The substances are burned at room temperature in a compound reaction with pure, gaseous oxygen according to reaction (II) to form gaseous CO_2 and liquid water.



The heat conversion $\Delta_r Q$ of reaction (II) is to be determined calorimetrically under isochoric conditions ($V = \text{const.}$). If ideal mixing behavior is postulated for the gas phase components, the following relationship results between the molar enthalpy of combustion $\Delta_c H_m$ of the reaction (II) and the isochoric heat conversion $\Delta_r Q$:

$$\Delta_c H_m = \left(\frac{\Delta_r Q}{\lambda} \right) + R \cdot T \cdot \sum_{i,(g)} \nu_i \quad (3)$$

With:

- $\Delta_c H_m$ = molar enthalpy of combustion [$\text{kcal} \cdot \text{mol}^{-1}$]¹
- $\Delta_r Q$ = heat of combustion [kcal]
- λ = throughput: 1 [mol]²
- R = universal gas constant: $1.9859 \cdot 10^{-3}$ [$\text{kcal} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$]
- T = ambient temperature [K]
- $\sum_{i,(g)} \nu_i$ = sum of stoichiometric coefficients of **gaseous** educts / products []

Equation (3) provides the basis for determining $\Delta_c H_m$ from the calorimetrically measured isochoric heat conversion $\Delta_r Q$. How can $\Delta_c H_m$ be converted to standard conditions $\Delta_c H_m^0$? With the experimental conditions given below, is the approximation $\Delta_c H_m \approx \Delta_c H_m^0$ justifiable? **These questions are to be answered along with the exercises (see chapter 6)!**

¹ Index “c” in $\Delta_c H_m$ indicates a combustion (Index “r” – as in “reaction” – would be a valid index, too).

Index “m” in $\Delta_c H_m$ means “molar”, indicating the normalization to the amount of substance (in moles).

² The throughput λ indicates consumption (in moles) of any reactant equivalent to the stoichiometric coefficient ν given in the chemical equation – e. g. 1 mole of benzoic acid and 7.5 moles of O_2 in reaction (III). To avoid complications, the chemical equation should always be written so that the stoichiometric coefficient ν of the “substance of interest” is 1 – e. g. $\nu_{\text{benzoic acid}} = 1$ in reaction (III).

2.3 Bomb calorimetry

The basic design of a bomb calorimeter is shown schematically in Figure 1.

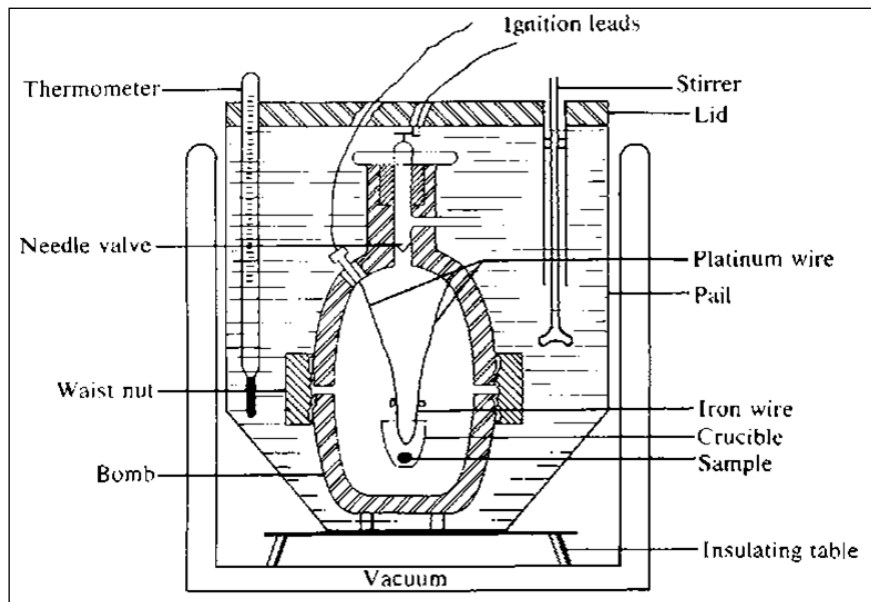


Figure 1: Detailed cross section of a general-purpose bomb calorimeter for the determination of combustion heats (from [2], p. 86).

As shown in

Figure 1, the substance to be burned is pressed into a pellet and put inside the calorimeter bomb filled with O_2 ($p = 30$ bar). The bomb is placed in a calorimeter vessel filled with water at a known initial temperature. The vessel is insulated from the outside and equipped with a stirrer and a digital thermometer. This setup allows for a rapid exchange of heat between the calorimeter bomb and the surrounding water jacket.

The reaction is electrically ignited using an ignition wire (usually made of iron) that contacts the sample pellet. The wire is partially oxidized during combustion. The total heat of reaction converted during the reaction $\Delta_r Q$ is exchanged with the water jacket and leads to a temperature change ΔT_X in the calorimeter vessel. ΔT_X is linked to $\Delta_r Q$ via the (unknown) heat capacity C of the total system as follows:

$$\Delta T_X = \left(\frac{\Delta_r Q(P) + \Delta_r Q(Fe)}{C} \right) \quad (4)$$

With ΔT_X = Temperature change in the reaction vessel caused by the reaction [K]
 $\Delta_r Q(Fe)$ = Heat of reaction generated by the combustion of the (iron) ignition wire [kcal]
 $\Delta_r Q(P)$ = Heat of reaction generated by the combustion of substance P [kcal]
 C = Heat capacity of the calorimeter system [kcal·K⁻¹]

The heat capacity C is determined by burning a known mass m_E of a calibration substance R for which the molar enthalpy of combustion $\Delta_c H_m(E)$ is known. The oxidation of benzoic acid (III) is usually used for this purpose:



With $\sum_{i,(g)} \nu_i = -\frac{1}{2}$

it then follows from the combination of eq. (3) and eq. (4), taking into account the relationship between $\Delta_r Q(Fe)$ and the specific heat of combustion $\Delta_c U(Fe)$ of the iron:

$$C = \frac{1}{\Delta T_X} \cdot \left[\frac{m(R)}{M(R)} \cdot (\Delta_c H_m(R) - R \cdot T \cdot \sum_{i,(g)} \nu_i) + \Delta m(Fe) \cdot \Delta_c U(Fe) \right] \quad (5)$$

With:

- C = heat capacity of the calorimeter system [kcal·K⁻¹]
- ΔT_X = measured temperature change [K]
- $m(R)$ = burned mass of reference substance (benzoic acid) [g]
- $M(R)$ = molar mass of burned reference substance [g·mol⁻¹]
(for benzoic acid: 122.12 g·mol⁻¹)
- $\Delta_c H_m(R)$ = molar enthalpy of combustion of reference substance [kcal·mol⁻¹]
(for benzoic acid: 771.603 kcal·mol⁻¹)
- R = universal gas constant: $1.9859 \cdot 10^{-3}$ [kcal·mol⁻¹·K⁻¹]
- T = ambient temperature [K]
- $\sum_{i,(g)} \nu_i$ = sum of stoichiometric coefficients of **gaseous** educts / products []
- $\Delta m(Fe)$ = combusted mass of (iron) ignition wire [g]
- $\Delta_c U(Fe)$ = specific heat of combustion of iron,
 ≈ 1.4 [kcal·g⁻¹] (see packaging of ignition wire)

For the molar enthalpy of combustion of the reaction of interest (II), the result is then analogous with known C :

$$\Delta_c H_m(P) = \frac{M(P)}{m(P)} \cdot [C \cdot \Delta T_X - \Delta m(Fe) \cdot \Delta_c U(Fe)] + R \cdot T \cdot \sum_{i,(g)} \nu_i \quad (6)$$

With

- $\Delta_c H_m(P)$ = molar enthalpy of combustion of sample substance P [kcal·mol⁻¹]
- $m(P)$ = combusted mass of substance P [g]
- $M(P)$ = molar mass of substance P [g·mol⁻¹]

2.4 Evaluation of Temperature-time curves

In the experimental determination of ΔT_X , the sluggish heat transfer dynamics of the calorimeter and any non-negligible heat exchange between the calorimeter vessel and the environment

usually result in a time-dependent temperature profile as shown in

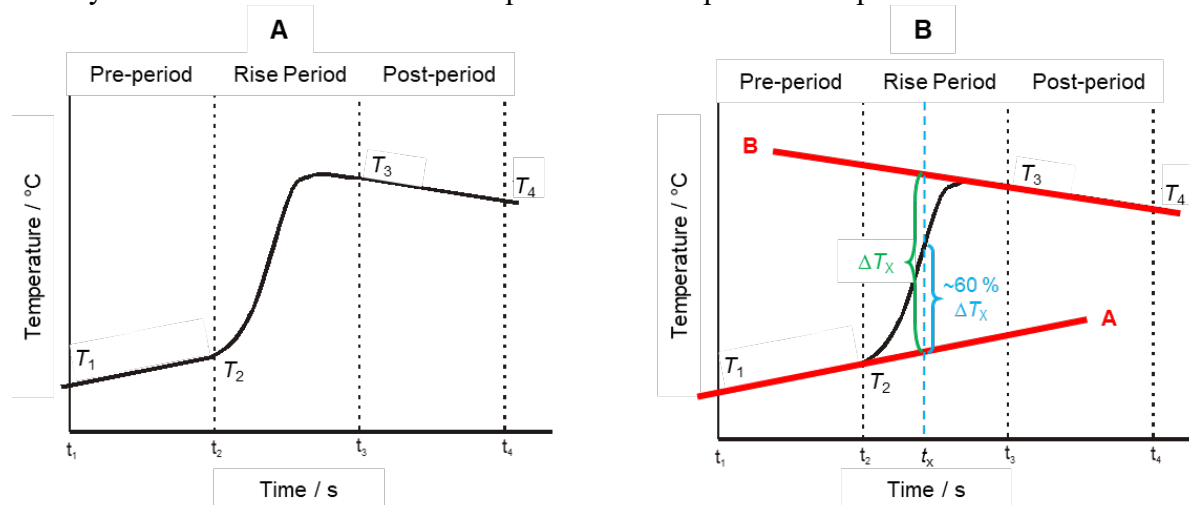


Figure 2A. The heat exchange with the environment leads to different temperature profiles in the measuring period before and after the reaction. The temperature rise associated with the reaction of interest ΔT_x can be most accurately determined by a procedure presented by the Bureau of Standards (nowadays NIST) in 1915 [3].

The temperature profiles of the pre- and post-period are linearly extrapolated, and their respective slopes are determined by equations (7) & (8). This is illustrated in

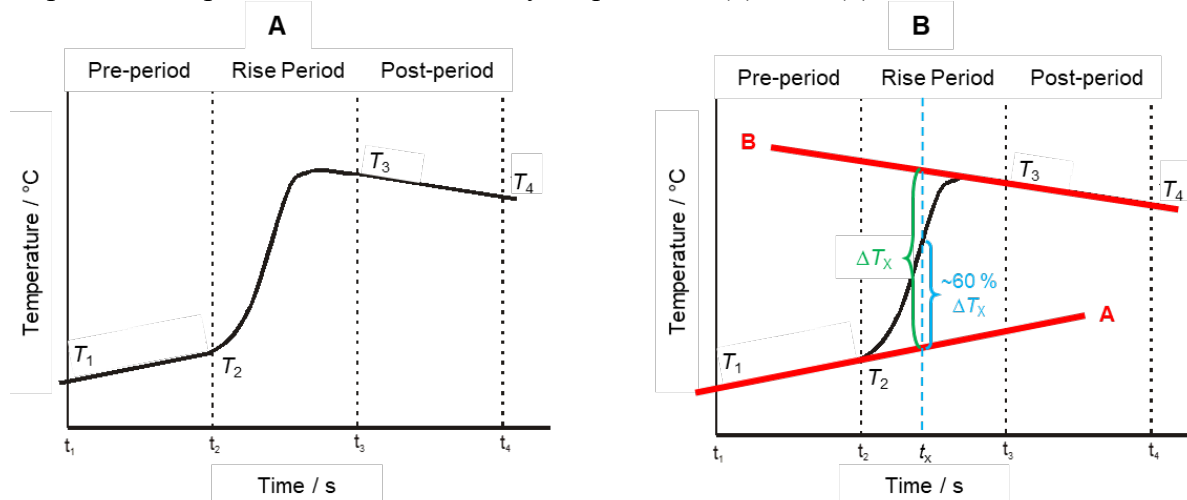


Figure 2B by the red lines labeled A and B, respectively.

$$r_A = \frac{T_2 - T_1}{t_2 - t_1} \quad (7)$$

$$r_B = \frac{T_4 - T_3}{t_4 - t_3} \quad (8)$$

With

- r_A = slope of line A (temperature change rate before ignition) [$\text{K} \cdot \text{s}^{-1}$]
- r_B = slope of line B (temperature change rate after ignition) [$\text{K} \cdot \text{s}^{-1}$]
- T_1 = temperature at the beginning of the experiment [K]
- T_2 = temperature immediately before ignition [K]
- T_3 = start of post-period
(onset of constant temperature change rate after rise period) [K]
- T_4 = end temperature (after a sufficient time to determine r_B) [K]

In most experiments, the mechanical stirring heats up the water in the calorimeter. Accordingly, r_A is usually positive. r_B is mostly negative, as the temperature usually decreases during the post-period.

Next, a line perpendicular to the abscissa is drawn, so that it intersects the temperature curve at approximately 60 % of its height between lines A & B. This is illustrated by the blue line in

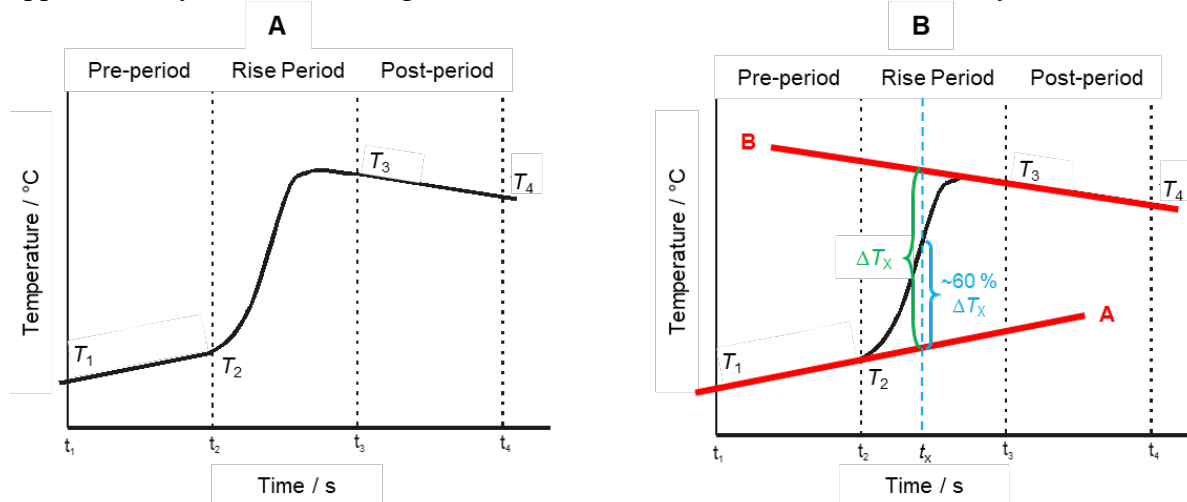


Figure 2B. It represents the time t_x .

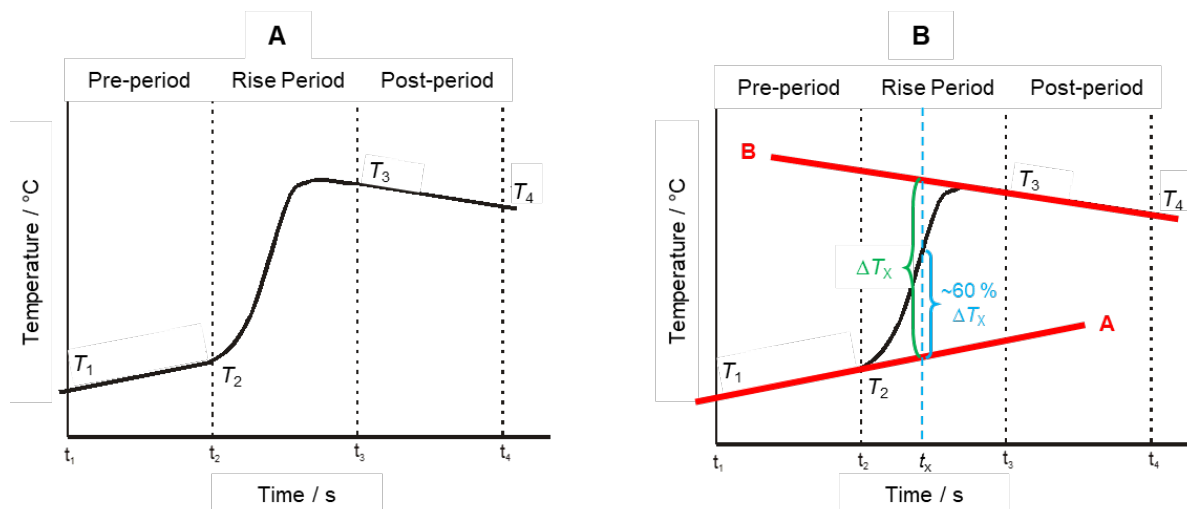


Figure 2: Temperature-time curves for exothermic calorimetric processes

A common curve shape for a fast reaction

B extrapolation of pre- and post-period to determine time t_x and temperature rise ΔT_x

Lastly, ΔT_x can be approximated by equation (9):

$$\Delta T_x = T_3 - T_2 - r_A \cdot (t_x - t_2) - r_B \cdot (t_3 - t_x) \quad (9)$$

For more details on the mathematical background of this graphical evaluation method, please refer to the appendix.

3. Equipment

For the determination of $\Delta_r Q$, a PARR 1341 bomb calorimeter is used. Detailed instructions on the operation of the bomb calorimeter can be found in Parr's user manuals 204M, 205M and 545M, which are enclosed with the experimental material (physically as well as on ILIAS). Please take special note of the following things when handling the equipment:

- Neither the lid of the calorimeter vessel nor the lid of the oxygen combustion vessel (bomb) should be placed on the working surface due to risk of contamination and/or damage. Special stands with a large or small ring are provided which allow the two lids to be safely placed while operating on other parts.
- The temperature sensor should be immersed to approximately the same depth as the stirrer. When inserting the sensor and adjusting the immersion depth, the support screw should never be fully loosened, as the ferrules get lost very quickly.

4. Experimental approach

The entirety of the experimental procedure is explained comprehensively in a video on ILIAS. Please watch it beforehand. The explanations below are meant to be used as guidelines while performing the experiment.

4.1 Experimental preparations

The oval water container is filled with exactly $2.0000 \text{ kg} \pm 0.5 \text{ g}$ (use the balance!) of deionized water and placed in the calorimeter vessel. The mass of the water must be measured precisely for each experiment so that the heat capacity C of the system remains constant.

The ignition unit (2901EE) is connected to the outside of the calorimeter vessel. Make sure that the two cables are connected to "common" and "10 cm".

Several pastilles (each approx. 900 - 1,000 mg) of the "calibration substance" benzoic acid are pressed. **The displayed "pressure" should not exceed 10 t!** Alternatively, a pastille from Parr (3415) can be used.

4.2 Determination of the calorimeter's heat capacity

A combustion capsule (43AS) is weighed using an analytical balance. To determine the mass of the pastille, it is directly weighed in the combustion capsule. The combustion capsule with the sample is inserted into the holder integrated in the lid of the combustion vessel. Approximately 10 cm of the ignition wire (45C10) are cut off and also weighed accurately. The ends of the ignition wire are threaded through the eyelets at the bottom of the two ignition electrodes. By lowering the caps, the ignition wire is fixed to the electrodes. The wire is then bent so that it makes **permanent contact with the sample** pastille. Also, the wire should **not touch the combustion capsule**. The lid is carefully re-mounted on the combustion vessel so that the ignition wire does not lose contact with the pastille.

The O_2 filling connection (1825) is used to fill the combustion vessel with oxygen. After connecting the pressure gauge to the oxygen tank (tighten well!), the control valve (black knob) is closed completely (turn clockwise until it stops). Then the main valve of the pressure bottle is opened (no more than $\frac{1}{4}$ -turn). The input gauge (small) will now indicate the pressure in the bottle, while the output gauge (large) should remain at 0 bar. The outlet of the O_2 filling connection is slipped over the plug connection on the combustion vessel and it is ensured that the outlet valve on the combustion vessel's lid is shut. Then, the pressure in the combustion vessel is built up by carefully opening the control valve until the output pressure gauge indicates 30 bar. The filling connection can be removed from the combustion vessel and the remaining pressure in the filling connection is released by pressing down the black lever on the O_2 filling connection (the display on the output gauge returns to 0 bar).

The combustion vessel should be filled with oxygen a total of 3 times to remove other gases from the system. The gas in the combustion vessel can be vented by opening the outlet valve on its lid.

The combustion vessel is partially immersed in the water bath by using the gripping tongs (421A). Before the combustion vessel is completely immersed, the two ignition cables inside the calorimeter vessel are connected to the combustion vessel's lid. After complete immersion, bubbles will escape from the lid for approximately 60 s. After that, no bubbles should escape from the vessel anymore. Otherwise, it must be checked for leaks and refilled. In addition, as little water as possible should be lost - shake out water from the gripping tongs.

At this point, prepare notebook material or a laptop to record the measured values. The Excel file "raw data_(group number)_FS(year).xls" (on ILIAS) serves as a template. Also, it might be advisable to record the temperature sensor display on video, as the temperature values change rather quickly.

Now the lid of the calorimeter vessel is put on and the drive belt is attached to the motor. After starting the motor, the digital thermometer is turned on and the starting temperature is noted. The temperature is recorded every 60 s for 6 minutes. At 06:00 min, the ignition button on the ignition unit is pressed for 5 s. The red lamp should light up for a short time³. During the next 6 min the temperature is read and noted every 10 s. During the post period (5 - 7 min), the temperature is again read every 60 s. The difference between the last two measurements should be $< 0.02\text{ }^{\circ}\text{C}$. The excel sheet „Calorimetry_raw data_(group number)_FS(year)“ (on ILIAS) may be used to note the raw data. Alternatively, the recorded video can be used to convert the digital data of the thermocouple using optical character recognition. See source [4] for a link to the corresponding application.

After completion of the measurement, the combustion vessel is carefully removed from the calorimeter vessel using the gripping tongs. The pressure is released by opening the outlet valve on the combustion vessel's lid. The mass of the remaining ignition wire and any unburned sample substance is measured. From these measurements, the parameters m_E and $\Delta m(\text{Fe})$ in equation (5) are determined.

ΔT_X is determined according to the graphic analysis shown in

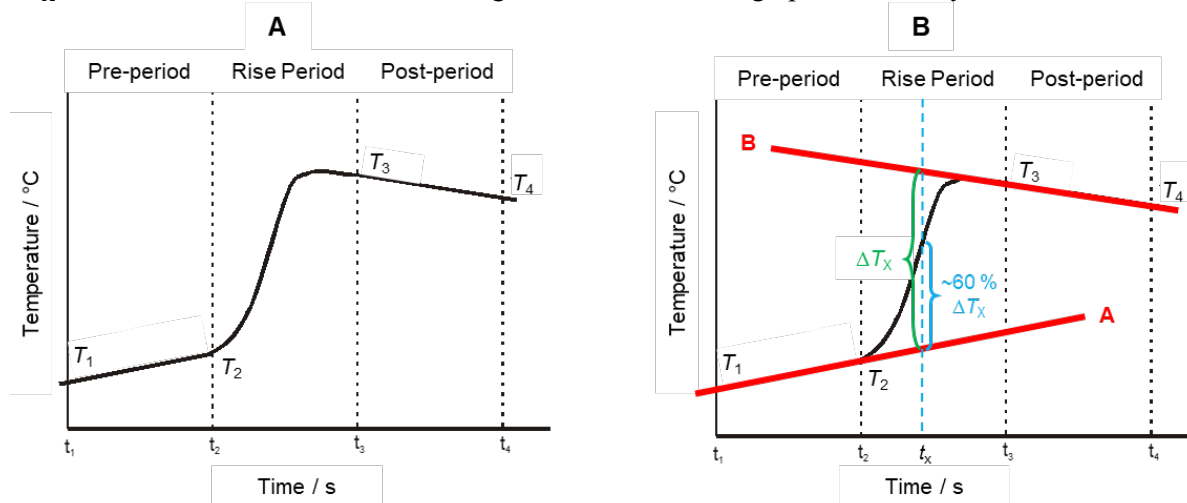


Figure 2. Equation (5) is used to determine the heat capacity C of the calorimeter using the literature value for the enthalpy of combustion of benzoic acid $\Delta_c H_m(\text{R}) = 772.12\text{ kcal/mol}$ ($T = 298\text{ K}$).

Chemistry and biochemistry students only:

Repeat the determination of the system's heat capacity in order to determine the measurement uncertainty of the obtained value.

4.3 Determination of molar enthalpy of combustion

After determining the heat capacity of system C , the measurements described above are carried out analogously with pastilles of the sample substances (maleic acid / fumaric acid). The pressed pills should

³ If the lamp has not turned off after 5 s, the sample has not been ignited because the ignition wires have moved away from the pastille. In this case, the measurement must be aborted and the system prepared again.

also have a mass of 900 – 1'000 mg. From the measured temperature rise ΔT_X and heat capacity C , the molar enthalpy of combustion of the sample substances $\Delta_c H_m(P)$ are determined using equation (6).

Chemistry and biochemistry students only:

Measure duplicates of each substance.

5. Data evaluation

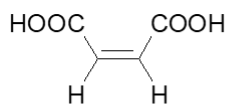
Plot the raw data with an appropriate software (*e. g.* MS Excel) and follow the instructions given in chapter 3. Alternatively, graph paper (to be found in the laboratory) may be used. Compare your results with literature values.

Chemistry and biochemistry students only:

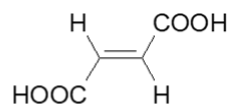
For each substance, calculate the mean value, the standard deviation.

6. Exercises

- Answer the questions from the text
- What is the rationale behind determining t_x via the 60 % intersection of ΔT_x ?
Hint: Read source [3], p. 230 - 233
- Calculate the enthalpies of combustion of maleic and fumaric acid
$$\text{HOOC}-(\text{CH})_2-\text{COOH} + 3 \text{ O}_2 \rightarrow 2 \text{ H}_2\text{O} + 4 \text{ CO}_2$$
- Discuss the differences between the measured enthalpies of combustion. Why is the enthalpy of combustion of glutaric acid higher than the ones of maleic and fumaric acid?
- How can the enthalpy of transformation of the reaction “maleic acid $\rightarrow$ fumaric acid” be determined?



maleic acid



fumaric acid

7. Literature

- [1] P. W. Atkins, J. de Paula and J. Keeler, *Physical Chemistry*, 11th edition, Oxford University Press, Oxford **2018**, chapters 2A.4 – 2C.3
- [2] F. A. Bettelheim, *Experimental Physical Chemistry*, 9th edition, W. B. Saunders Company, Philadelphia **1971**, chapter 7
- [3] H. C. Dickinson, *Combustion calorimetry and the heats of combustion of cane sugar, benzoic acid, and naphthalene*, NBS Bulletin **1915**, 11(2), p. 189-257,
<http://dx.doi.org/10.6028/bulletin.252>
- [4] R. Schmidt, DCBP Universität Bern, Python script *read-characters-from-video*,
<https://github.com/romanschmidxyz/read-characters-from-video>, 03.02.2026

Appendix

The loss of heat from the calorimeter to the jacket and the addition of heat through stirring during the experiment must be taken into account when determining the temperature change ΔT_X in the calorimeter vessel. Four main modes of transport are involved: air conduction, convection, radiation, and evaporation. Observations have shown that Newton's law of cooling can be assumed to account for this heat loss which yields the following differential equation⁴:

$$\frac{dT}{dt} = f(t) - k \cdot (T(t) - T_{\text{env.}}) \quad (11)$$

With $\frac{dT}{dt}$ = rate of temperature change of the system [$\text{K} \cdot \text{s}^{-1}$]
 $f(t)$ = temperature change caused by sample combustion [$\text{K} \cdot \text{s}^{-1}$]
 k = constant [s^{-1}]
 $T(t)$ = temperature of the calorimeter [K]
 $T_{\text{env.}}$ = jacket temperature [K]

This equation can be solved to yield the following:

$$T = T_{\text{env.}} \cdot (1 - e^{-k \cdot t}) + \int_0^t f(t) dt \quad (12)$$

But since we do not know the exact contribution of $f(t)$ to the temperature increase throughout the experiment, we cannot calculate ΔT_X directly. However, a $T-t$ diagram as in Figure A1 can be used as it contains information about the behavior of the system without any influence of $f(t)$ before and after the reaction. t_i is the start and t_f the end of the rising period.⁵ A time t_m with the following property can be calculated:

$$\int_{t_i}^{t_f} (T - T_i) dt = \int_{t_i}^{t_m} (T_i - T_j) dt + \int_{t_m}^{t_f} (T_f - T_j) dt \quad (13)$$

With area $AA'E$ being equal to area $DD'E$ in Figure A1. Thus:

$$\int_{t_i}^{t_m} (T - T_i) dt = \int_{t_m}^{t_f} (T_f - T) dt \quad (14)$$

The pre-reaction and post-reaction curves are linearly extrapolated and ΔT_X is calculated as the temperature difference between the extrapolated curves at t_m .

⁴ For clarity: The rate of change of temperature $\frac{dT}{dt}$ is proportional to the heat transfer $\frac{dQ}{dt}$

⁵ Note that the nomenclature for these times has changed in comparison to chapter 2.4.

For example, t_i in Figure A1 corresponds to t_2 in Figure 2A.

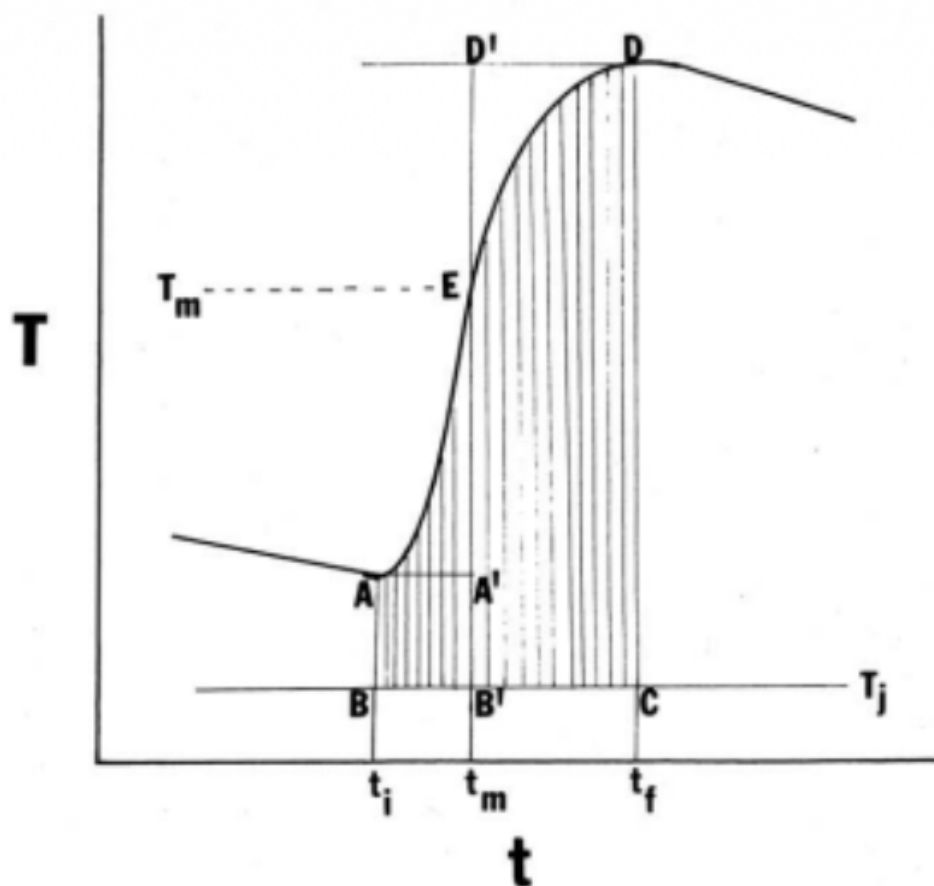


Figure A1. T - t diagram for combustion reaction with calorimetry lag [3].